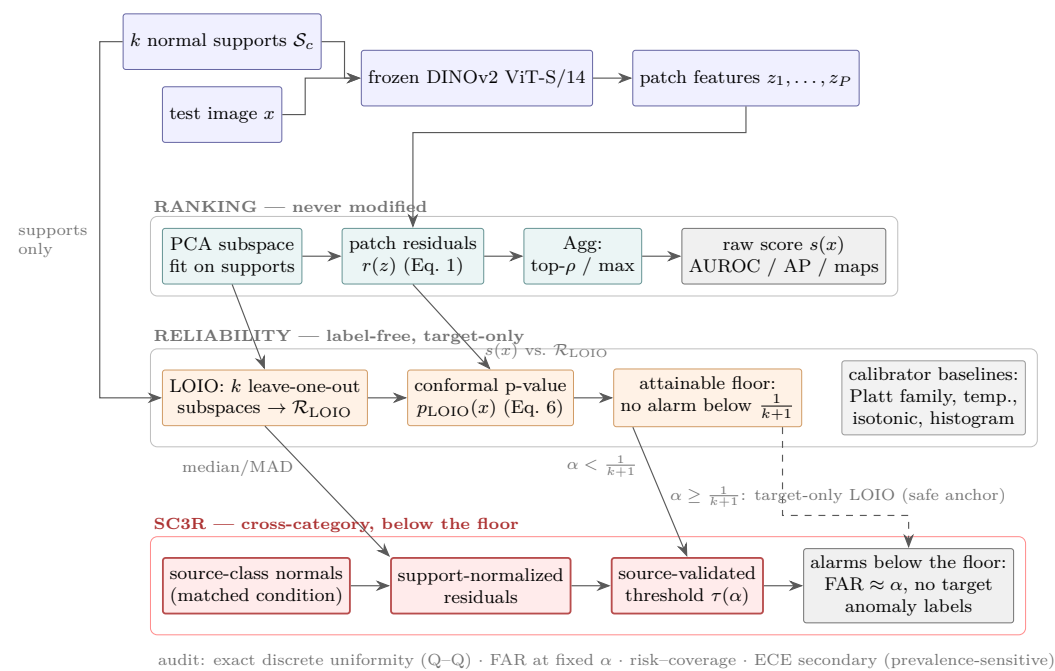


Graphical Abstract

Conformal Reliability Routing for Low-Storage Few-Shot Industrial Anomaly Detection

Given Family



Highlights

Conformal Reliability Routing for Low-Storage Few-Shot Industrial Anomaly Detection

Given Family

- A k -shot conformal anomaly alarm is structurally silent below $\alpha = 1/(k+1)$
- Exact discrete uniformity tests localize when the false-alarm promise inverts
- SC3R pools cross-category normals to alarm below the floor without target labels
- Pre-registered gate passes on 15 MVTec classes and replicates on 12 VisA classes
- LOIO conformal beats Platt, temperature, isotonic, and histogram calibrators

Conformal Reliability Routing for Low-Storage Few-Shot Industrial Anomaly Detection

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ABSTRACT

What can an alarm calibrated on only k normal examples of a learned representation actually guarantee, and where does it structurally fail? We study this question for few-shot anomaly detectors built on frozen foundation features, such as DINOv2 patch embeddings scored with memory banks or subspace residuals: they rank industrial defects well, but their raw scores are not calibrated probabilities and degrade under low-shot, dataset, and corruption shift. We present *Conformal Reliability Routing* (CRR), a decoupled framework that keeps a low-storage PCA/subspace residual (under 0.6MB per class) as the anomaly ranking score and adds a separate label-free reliability layer built from leave-one-image-out (LOIO) conformal p-values computed only from the k normal support images. Because a conformal p-value promises false-alarm control rather than a posterior probability, we evaluate it with conformal-native evidence—false-alarm rates at fixed levels, exact discrete uniformity tests, attainable-alpha analysis, and risk-coverage curves—with calibration error as a secondary, prevalence-sensitive metric. On full corruption benchmarks over all 12 VisA and all 15 MVTec classes, the LOIO route improves calibration over Platt-family and standard scalar calibrators under paired Wilcoxon tests while leaving the ranking unchanged; uniformity tests show target-only p-values are conservative on VisA, anti-conservative on corrupted MVTec, and structurally silent below $\alpha = 1/(k+1)$. To alarm below this floor, we introduce SC3R, which pools normal images from other categories and source-validates alarm thresholds under matched conditions. Against a pre-registered decision gate, SC3R tracks nominal false-alarm rates almost exactly on all 15 MVTec classes with bootstrap-supported power gains under every corruption condition, replicates on all 12 VisA classes, and transfers conservatively with cross-dataset source archives. CRR is a calibrated, low-storage reliability layer that attaches to an existing learned detector; beyond inspection, it characterizes when few-shot conformal alarms on learned features can and cannot be trusted.

1. Introduction

Industrial anomaly detection (AD) is typically deployed under an asymmetric data regime: normal examples are available, while abnormal examples are rare, diverse, and costly to annotate. This makes few-shot AD practically important. In this setting, an inspection system receives only k normal support images for a target category, often with $k \in \{1, 2, 4, 8\}$, and must detect both image-level and pixel-level defects at test time.

Recent methods have made this regime much more effective by using frozen visual foundation models. DINOv2 [1] patch features can be stored in a normal memory bank and queried by nearest-neighbor distances, as in PatchCore-style methods and AnomalyDINO [2, 3]. Alternatively, frozen patch features can be compressed into a normal PCA/subspace model, as in SubspaceAD [4]. These approaches provide strong anomaly ranking without training a large task-specific backbone.

However, ranking is not the same as reliable decision support. A raw nearest-neighbor distance or PCA residual can produce high AUROC while still being poorly calibrated: a score threshold may work on one class, but the same numerical score may not carry the same false-alarm meaning under another class, dataset, or corruption. This matters in industrial deployment because operators need to know not only which samples are suspicious, but also whether an alarm probability is trustworthy. Prior work has already shown that DINOv2-based few-shot AD can be poorly calibrated and adversarially fragile [5]. Reliability is therefore not a cosmetic post-processing issue; it is part of the deployment problem.

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We address this gap by separating anomaly ranking from reliability estimation. The resulting questions are about the learning system itself, not about any one factory: what false-alarm behavior can a decision rule calibrated on k examples of a frozen representation guarantee, at what resolution, and how does that guarantee fail when the representation’s inputs shift? The ranking path remains a frozen DINOv2 PCA/subspace detector: patch residuals are aggregated into a raw anomaly score used for AUROC, AP, and localization. A separate reliability path maps this evidence into calibrated or probability-like outputs. We call the resulting framework *Conformal Reliability Routing* (CRR). The main CRR route uses leave-one-image-out (LOIO) conformal calibration over the few normal support images. Each support image is scored as if held out, producing normal nonconformity values; a test image then receives a conformal p-value and a reliability score $1 - p$. This view is not a supervised anomaly posterior, but it is statistically interpretable and derived without target anomaly labels.

Each ingredient we build on—DINOv2 features, PCA residuals, Platt-family calibration, and conformal anomaly detection—has an established literature, and our contribution begins after them. What that literature leaves open is a structural question about the few-shot regime itself: a conformal alarm calibrated on k support images cannot fire below $\alpha = 1/(k+1)$ at all, and under corruption shift its false-alarm promise can silently invert. The central contribution of this paper is that pair of problem and mechanism—making the resolution floor and its failure directions exact and checkable, and then introducing a source-validated cross-category scheme (SC3R) that alarms below the floor with controlled false alarms and no target anomaly labels. The decoupled detector and the calibration benchmark around them are the supporting substrate: a protocol under which these claims can be tested without conflating ranking and reliability.

The lesson for learning systems is general even though the testbed is industrial: the reliability of a few-shot alarm is limited by an exact resolution floor of its calibration set, its failure direction under shift is measurable, and both can be repaired with data the deployment already has. The paper makes four contributions, the first two central:

- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with k supports no alarm can fire below $\alpha = 1/(k+1)$, and empirical false-alarm rates must be read against the attainable grid, not the nominal level. Exact discrete uniformity tests of normal p-values then localize when and in which direction the false-alarm promise fails under corruption shift—conservative on VisA, anti-conservative on corrupted MVTec.
- We introduce SC3R, a source-validated cross-category thresholding scheme that pools normal images from other categories to unlock operating points below the target-only attainable-alpha floor. Under a pre-registered decision gate on all 15 MVTec classes, matched-condition SC3R tracks nominal false-alarm rates almost exactly with substantial power where the target-only anchor is structurally silent, and hierarchical class-seed bootstrap intervals for its power gain exclude zero for every corruption condition. The result replicates on all 12 VisA classes, and cross-dataset source archives (MVTec sources for VisA targets) transfer conservatively: reduced power with false alarms well below nominal.
- Supporting these claims, we formulate a decoupled few-shot AD pipeline in which frozen DINOv2 PCA/subspace residuals provide anomaly ranking while a separate label-free reliability layer—vector/shift-aware calibration and LOIO conformal p-values [6] computed only from the k normal supports—provides decision signals, with full-benchmark corruption evidence on all 12 VisA and all 15 MVTec classes that the conformal route improves calibration over Platt-family calibrators and over standard scalar calibrators (temperature scaling, isotonic regression, histogram binning) under paired Wilcoxon tests while preserving the raw ranking.
- We report accuracy-storage, transfer calibration, pixel-level, robustness, prevalence-stress, and routing diagnostics, including negative findings that prevent overclaiming: CRR is not MVTec AUROC SOTA, is not adversarially robust, and ECE on conformal-derived scores is strongly prevalence-sensitive.

The remainder of this paper is organized as follows. Section 2 positions the work against few-shot detectors, calibration, and conformal anomaly detection. Section 3 presents the decoupled pipeline, the LOIO conformal route, the attainable-alpha analysis, and SC3R. Section 4 describes datasets, protocols, and claim rules. Section 5 reports calibration comparisons, uniformity tests, operational false-alarm control, SC3R evaluations with their pre-registered gate, and diagnostics. Section 6 discusses limitations, and Section 7 concludes.

2. Related Work

2.1. Few-Shot Industrial Anomaly Detection

Classical industrial AD often assumes normal-only training data and detects deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative families such as PaDiM, CutPaste, and DRAEM provide important context for this literature [7–9]. PatchCore [2] made memory-bank patch retrieval a strong baseline by storing representative normal features and scoring test patches by nearest-neighbor distance. AnomalyDINO [3] updates this paradigm for few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has also produced registration-based adaptation (RegAD [10]), graph-structured memory (GraphCore [11]), fast feature reconstruction (FastRecon [12]), vision–language zero-/few-shot scoring (WinCLIP [13]; CLIP–SAM collaboration [14] and cross-modal prompt regularization [15] continue this line), and in-context residual learning (InCTRL [16]). These methods are strong rankers, but the reference state can remain nontrivial at inference time and the raw score scale is not a calibrated probability; calibration under shift is also known to degrade for deep models generally [17]. Our work is complementary: it does not compete on ranking but asks how any such few-shot detector should expose decision reliability.

2.2. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [4] shows that frozen DINOv2 patch features and PCA/subspace residuals are already strong for few-shot anomaly ranking; the ranking mechanism is therefore settled prior art. Our contribution starts after the residual is computed: how should a low-storage subspace detector expose reliability under low-shot and shift, and what can its alarms guarantee?

2.3. Calibration and Robustness in Few-Shot AD

Calibration asks whether predicted confidence agrees with empirical correctness [18, 19]. Standard post-hoc calibrators include Platt scaling [18], temperature scaling [19], histogram binning [20], and isotonic regression [21]; we compare against all of them under a shared label-free protocol. In anomaly detection this is subtle because labels are sparse, anomaly types are open-ended, and the score distribution changes across classes. Reliability analyses of DINOv2-based few-shot AD report poor ECE and strong FGSM fragility [5]. We therefore evaluate ECE, Brier score, and NLL separately from AUROC/AP, and report adversarial results, including FGSM-style perturbations [22], as diagnostics rather than robustness improvements.

2.4. Conformal Prediction and Reliability Routing

Conformal prediction converts nonconformity scores into p-values under exchangeability assumptions [23–25]. This is attractive for normal-only few-shot AD because support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection itself is established: leave-one-out, bootstrap, and cross-conformal anomaly detectors have been analyzed directly [6], weighted conformal detection in low-data regimes with resolution and effective-sample-size analyses exists [26], general nonconformity toolkits are available [27], and few-shot conformal prediction with auxiliary tasks addresses the small-calibration-set problem [28]. Conformal AD, LOIO calibration, and weighted conformal p-values are thus established mechanisms, and our contribution begins where they stop in the few-shot regime: at the resolution floor $1/(k+1)$ that no target-only k -shot conformal alarm can cross. We make that floor and its corruption-shift failure directions exact and checkable, and introduce source-validated cross-category thresholding that operates below the floor without target anomaly labels. Gated-expert models, including SAGE-style shared/specialized routing [29], motivate the broader idea of choosing conservative or shift-specialized reliability paths. CRR uses this routing lens, but its main reported route is intentionally simple: fixed LOIO conformal reliability. Learned or weighted gates are treated as ablations unless they improve reliability without test-label expert selection.

3. Method

3.1. Problem Setup

For each category c , we observe a support set $S_c = \{x_i\}_{i=1}^k$ containing only normal images. The test set contains normal and anomalous images, but anomaly labels are used only for evaluation, not for fitting the main detector or calibrator. The goal is to produce: (i) a raw anomaly ranking score $s(x)$, (ii) a probability-like reliability score $\hat{p}(y = 1 \mid x)$ or conformal anomaly confidence, and (iii) optional pixel-level maps. Figure 1 gives an overview: a

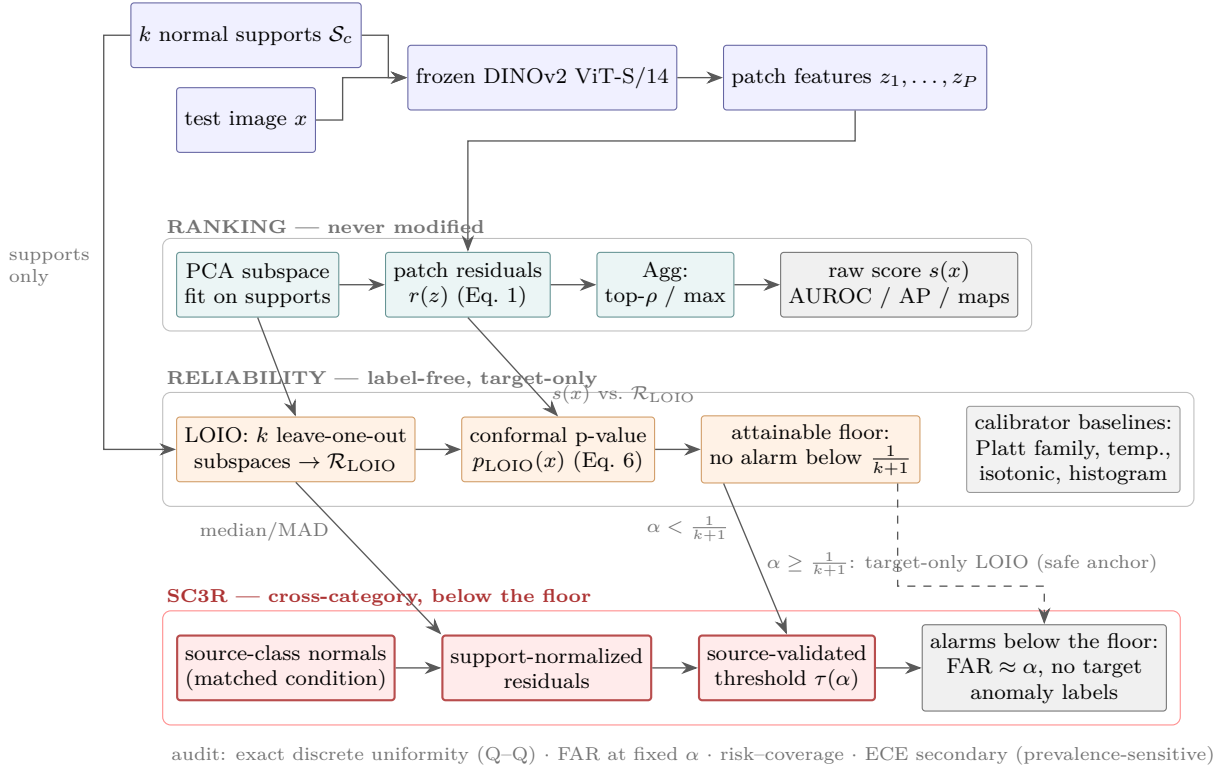


Figure 1: CRR overview. A frozen DINOv2 backbone feeds two decoupled paths: the *ranking* lane (PCA residuals, never modified; used for AUROC/AP and localization) and the label-free *reliability* lane (LOIO conformal p-values from the k supports, with Platt-family and standard scalar calibrators as baselines). Target-only alarms are structurally silent below the attainable floor $1/(k+1)$; the SC3R lane crosses it by pooling source-class normals under matched conditions and source-validating the threshold, deferring to the target-only anchor at levels the anchor can reach. The bottom strip lists the conformal-native audit used throughout.

frozen backbone feeds a ranking lane that is never modified, a label-free reliability lane whose conformal alarms hit the attainable-alpha floor, and the SC3R lane that crosses the floor with source-class normals.

3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i . A PCA/subspace model is fit on support patches. With support mean μ and retained principal directions $U \in \mathbb{R}^{d \times m}$, the patch residual is

$$r(z) = \left\| (z - \mu) - UU^T(z - \mu) \right\|_2. \quad (1)$$

Patch residuals form an anomaly map. The image-level ranking score is a high-residual aggregation,

$$s(x) = \text{Agg}_j r(z_j), \quad (2)$$

where Agg is the patch maximum in the clean accuracy-storage benchmark and the mean of the top-1% patch residuals ($\rho=0.01$) in the conformal pipeline; the top-fraction variant is a smoothed maximum that stabilizes the LOIO calibration scores. We state this explicitly because the two protocols use different aggregators, and Section 5 reports an aggregation-sensitivity ablation. Importantly, this raw score is the quantity used for AUROC and AP. Calibration routes do not change the ranking score.

3.3. Vector and Shift-Aware Calibration

The first reliability family is post-hoc logistic calibration. A scalar Platt scaler maps $s(x)$ to a probability, while Vector Platt uses a richer descriptor

$$\phi(x) = [s_{\text{pca}}(x), s_{\text{head}}(x), |\tilde{s}_{\text{pca}}(x) - \tilde{s}_{\text{head}}(x)|], \quad (3)$$

where s_{head} is a lightweight synthetic-anomaly head score and tildes denote normalized scores. The head is a small MLP ($384 \rightarrow 256 \rightarrow 1$, dropout 0.1) trained with binary cross-entropy to separate support normal patches from synthetically perturbed patches; a patch perturbation resamples a support patch and adds Gaussian noise plus a sign shift in feature space. Its image-level score is the maximum patch probability. The head participates only in the calibration descriptor $\phi(x)$, never in the anomaly ranking. The calibrated probability is

$$\hat{p}_\theta(x) = \sigma(w^\top \phi(x) + b). \quad (4)$$

Shift-Aware Platt augments $\phi(x)$ with five shift descriptors: the maximum and mean distance of test patches to the support feature center (scaled by the support 95th-percentile norm), the mean and standard deviation of patch residual scores, and a residual concentration ratio (max over mean). Weighted Platt reweights calibration samples by density-ratio estimates from a logistic domain classifier, following weighted conformal prediction under covariate shift [30]. These variants are useful for structured shifts but can over-adapt, motivating a conservative conformal route.

Calibrator training protocol and label source. All calibrators in the main protocol are trained without any real anomaly labels (mode `normal_synthetic`). The calibration set consists of the k support normal images (label 0) and an equal number of synthetic images (label 1) obtained by replacing a random 25% of a support image’s patch features with perturbed support patches as above. Vector-family calibrators fit a multivariate logistic model on $\phi(x)$ by gradient descent with L_2 regularization; the memory-bank and plain-subspace baselines, which expose no calibration descriptor, use scalar Platt on the raw score under the same synthetic protocol. The standard scalar calibrators evaluated in Section 5—temperature scaling [19], isotonic regression [21], and histogram binning [20]—are fit per class on exactly the same calibration set, mapping the raw score $s(x)$ to a probability; since raw residuals are not logits, temperature scaling is applied after centering at the midpoint of the two calibration class means, $\hat{p}(x) = \sigma((s(x) - c)/T)$, with only T learned. An `anomaly_val` mode that calibrates on 10% held-out real anomalies is reported only as an explicitly labeled upper bound. This symmetric, label-free protocol is what makes the LOIO-versus-calibrator comparison fair: every family sees exactly the same k support normals and no test information.

3.4. LOIO Conformal Reliability

For $k \geq 2$, CRR constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i , a subspace is fit on $\mathcal{S}_c \setminus \{x_i\}$ and the held-out image receives a residual score $s_{-i}(x_i)$. The resulting set

$$\mathcal{R}_{\text{LOIO}} = \{s_{-i}(x_i) : x_i \in \mathcal{S}_c\} \quad (5)$$

serves as normal nonconformity evidence. For a test image x , the conformal p-value is

$$p_{\text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{\text{LOIO}}} \mathbf{1}\{r \geq s(x)\}}{1 + |\mathcal{R}_{\text{LOIO}}|}. \quad (6)$$

We report $p_{\text{conf}}(x) = 1 - p_{\text{LOIO}}(x)$ as a probability-like anomaly reliability view. It should be interpreted as relative extremeness against held-out support normals, not as a fully supervised posterior probability. Weighted conformal replaces the uniform count by density-ratio weights [30] and reports effective sample size; empirically, it is less stable than LOIO on the main benchmark.

Exchangeability caveat. The construction above is deliberately asymmetric: each calibration residual $s_{-i}(x_i)$ comes from a subspace fit on $k-1$ images, while the test score $s(x)$ uses all k . The two statistics are therefore not exactly exchangeable, and Eq. (6) is a leave-one-out approximation in the spirit of cross-conformal and jackknife+ inference [6, 23, 31] rather than a split-conformal p-value with a formal finite-sample guarantee [32, 33]. We handle this empirically on two fronts. First, a matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces (pairing $s_{-i}(x)$ with r_{-i} , the jackknife+ pairing): it does not improve the operational false-alarm/power trade-off (Section 5), so the simpler construction is retained. Second, rather than assuming validity, we test it directly: Section 5 reports exact discrete-grid uniformity tests of the normal p-values, which quantify when and in which direction exchangeability fails under corruption shift.

3.5. Attainable Alpha and the Few-Shot Resolution Floor

Because $|\mathcal{R}_{\text{LOIO}}| = k$, the LOIO p-value is quantized on the grid $\{j/(k+1)\}_{j=1}^{k+1}$ and its smallest attainable value is $1/(k+1)$. Any nominal false-alarm level $\alpha < 1/(k+1)$ therefore cannot raise alarms at all: the detector is not merely conservative but structurally silent. At $k=4$ this floor is 0.2; at $k=8$ it is $1/9 \approx 0.111$. We make this resolution limit explicit with an attainable-alpha analysis that reports, for each nominal α , the nearest attainable grid point and the empirical false-alarm rate achieved there. This turns a commonly observed “few-shot conformal p-values are conservative” effect into an exact, checkable statement, and it defines the regime in which cross-category support is genuinely needed. The textbook remedy for this discreteness is the smoothed (randomized) conformal p-value, which tie-breaks with an independent uniform variable and is exactly $U(0, 1)$ under exchangeability; Section 5 evaluates it as a baseline and shows that randomization crosses the floor but inherits the target-only rule’s shift-induced bias, which is precisely what SC3R corrects.

3.6. Source-Conditioned Cross-Category Reliability Routing (SC3R)

The attainable-alpha floor motivates the main routing contribution. In multi-category factories, normal images from *other* categories (source classes) are plentiful even when the target category has only k normal supports. SC3R uses them in two steps that never touch target anomaly labels. First, target and source residuals are made comparable by support normalization: each raw score is centered and scaled by the robust median/MAD statistics of its own class’s LOIO support residuals. Second, an alarm threshold on conformal p-values is *source-validated*: for a candidate level α , held-out source classes are conformalized against the remaining source pool, and the largest p-value threshold whose empirical false-alarm rate on held-out source normals stays at or below α is selected. In the matched-condition mode, source images share the target’s corruption condition; in the clean-source mode, only clean source images are used. Pooling hundreds of source normals gives the p-value distribution the resolution that k target supports cannot: operating points below $1/(k+1)$ become attainable, while the source-validation step controls the induced false-alarm rate. A safe-anchor rule keeps target-only LOIO available: at levels at or above the target floor, where the anchor is well defined and strong, SC3R defers to it.

The routing component is inspired by the shared-versus-specialized principle in gated-expert systems: a conservative route should remain available, while shift-specialized routes can be used only when they improve reliability without harming the safe baseline. In our implementation, the conservative anchor is LOIO conformal reliability. Candidate experts include Vector Platt, Shift-Aware Platt, weighted Platt, LOIO conformal, and weighted conformal. We evaluate fixed routes, validation-ECE selection, and simple no-label rules based on shift descriptors and effective sample size. Because learned gates did not yet dominate fixed LOIO across all settings, the main paper reports the fixed LOIO route and treats gated variants as diagnostic ablations.

3.7. Selective Diagnostics

Beyond producing a calibrated alarm score, CRR records entropy, expert disagreement, conformal confidence, and effective sample size as selective-risk signals. These quantities are not used to change the anomaly ranking. Instead, they support deployment diagnostics such as abstention, risk-coverage analysis, and manual review of low-confidence alarms.

4. Experiments

4.1. Datasets and Few-Shot Splits

We evaluate on MVTec AD [34] and VisA [35]. MVTec AD contains industrial object and texture categories with pixel-level defect masks. VisA contains a broader set of industrial categories and is used as the main full-scale corruption and calibration-shift benchmark. For each class, support sets are sampled from normal training images with fixed seeds. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is used for clean accuracy-storage studies, and $k \in \{4, 8\}$ is used for the full conformal corruption study because LOIO conformal requires at least two support images and is most relevant in moderate few-shot settings.

4.2. Evaluation Protocols

We organize experiments into five protocol groups. First, the clean few-shot protocol evaluates image AUROC, AP, calibration, and storage on MVTec and VisA. Second, the accuracy-storage protocol compares PCA64 and PCA128 to test whether a small subspace increase improves accuracy without approaching memory-bank storage. Third, the

corruption reliability protocol evaluates Gaussian noise, blur, brightness/contrast shift, and JPEG compression at $k \in \{4, 8\}$ with at most 120 label-stratified images per case: on VisA across all 12 classes and five seeds (480 cases, 56,000 image rows), and on MVTec across all 15 classes and three seeds (360 cases, 37,464 image rows); VisA serves as the primary full-scale benchmark with the larger seed budget, while the MVTec replication uses three seeds for compute reasons, which its paired per-cell statistics account for. Fourth, representative routing protocols evaluate MVTec $\rightarrow$ VisA, VisA $\rightarrow$ MVTec, leave-one-class-out (LOCO), and within-dataset class splits. Fifth, the operational false-alarm protocol thresholds conformal p-values at nominal levels $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$ and reports false-alarm rate on normal images, detection power, alarm precision, and the attainable-alpha structure; the SC3R protocol additionally selects thresholds on held-out source-class normals with label-stratified random image sampling and a fixed sampling manifest, and is judged against a decision gate registered before the stratified run (nonzero power at $\alpha \leq 0.10$, mean false-alarm rate at most $\alpha+0.02$, no-harm at least 80%, no target anomaly labels, hierarchical class-seed bootstrap support).

4.3. Naming Convention

CRR denotes the overall framework (decoupled ranking plus reliability layer). The detector instantiations are named by their subspace configuration: CRR/PCA64 (default, also called CalibSubspaceHead in configuration files) and CRR/PCA128. Reliability views are named by their route: scalar/Vector/Shift-Aware/weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. Tables use these names consistently.

4.4. Baselines and Ablations

We compare against a controlled DINOv2 nearest-neighbor memory-bank baseline, a SubspaceAD-style PCA residual baseline, scalar Platt, Vector Platt, Shift-Aware Platt, weighted Platt, temperature scaling, isotonic regression, histogram binning, LOIO conformal, and weighted conformal. The controlled memory-bank implementation is not an official reproduction of PatchCore or AnomalyDINO; it holds the frozen DINOv2 feature pipeline and data split fixed to isolate the scoring and storage trade-off. In addition, we run the official AnomalyDINO code (ViT-S/14 at 448px, agnostic preprocessing) on both MVTec and VisA (official 1-cls split) for $k \in \{1, 4, 8\}$ with three seeds and report it as a separate accuracy reference. As the vision-language representative we include WinCLIP [13], but only as reported numbers: no official implementation is released, and when we audited the leading community reimplementations under its own protocol (ViT-B-16+240, LAION-400M; zero-shot and 1-shot with its fixed support lists, three runs), it reproduced 71.7/77.4 mean image AUROC on MVTec and 66.0/70.0 on VisA against the reported 91.8/93.1 and 78.1/83.8—a roughly 20-point gap that the reimplementations’ own authors also report and that we cannot attribute. This is precisely why our claim rules separate audited rows from reported rows, and WinCLIP therefore enters Table 1 as a cited reference, not an audited baseline. Official SubspaceAD representative runs are included as a novelty guardrail. Their strong performance confirms that frozen DINOv2 subspace residuals are an existing competitive ranking mechanism; the residual is inherited, and CRR’s contribution begins at the reliability of the alarms built on it.

4.5. Metrics

We separate ranking metrics from reliability metrics. AUROC and AP are computed from the raw PCA/subspace residual score $s(x)$. Pixel-level localization is measured by pixel AUROC and AU-PRO when masks are available. Reliability is measured operationally by empirical false-alarm rate at fixed α , detection power, alarm precision, discrete uniformity of normal p-values, and risk-coverage curves; expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) are reported as secondary metrics using the selected reliability probability or probability-like conformal view. Storage is measured as retained reference/model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate; the latter is used only as a fragility diagnostic.

4.6. Reproducibility and Claim Rules

All experiments run on a single NVIDIA RTX 5090 GPU with PyTorch in a pinned conda environment; DINOv2 patch features are extracted once and cached in fp32, so every calibration, conformal, and SC3R analysis is deterministic given the cached features and the stated seeds. All runs are indexed by dataset, class, k , seed, model variant, calibration mode, and corruption/attack setting, and per-image prediction tables with sampling manifests are retained for every reported number (see the data and code availability statements). The core rule is that reliability routes cannot claim

Table 1

Clean image-level performance and storage (means over classes and 5 seeds; official rows: 3 seeds). Best accuracy within each dataset/ k block is bold. Official AnomalyDINO rows are reproductions with the released code under its own protocol (ECE/storage not audited, left blank). WinCLIP rows are the numbers reported in [13]: no official implementation is released, and our audit of the leading community reimplementation falls about 20 AUROC points short of these numbers (see text), so WinCLIP is cited rather than audited. Controlled-NN ECE is diagnostic, not competitive (degenerate constant probability; see text), and CRR ECE at $k=1$ uses the vector-Platt view since LOIO requires $k \geq 2$.

Dataset	k	Method	AUROC $\uparrow$	AP $\uparrow$	ECE $\downarrow$	MB $\downarrow$
MVTec	1	Official AnomalyDINO	0.965	0.982	–	–
		WinCLIP (reported)	0.931	–	–	–
		Controlled DINOv2 NN	0.914	0.952	0.277	2.005
		CRR / CalibSubspace	0.904	0.951	0.274	0.472
	4	Official AnomalyDINO	0.976	0.984	–	–
		WinCLIP (reported)	0.952	–	–	–
		Controlled DINOv2 NN	0.942	0.967	0.693	6.000
		CRR / CalibSubspace	0.937	0.967	0.214	0.472
	8	Official AnomalyDINO	0.980	0.990	–	–
		Controlled DINOv2 NN	0.948	0.970	0.693	6.000
		CRR / CalibSubspace	0.945	0.972	0.154	0.472
VisA	1	Official AnomalyDINO	0.857	0.866	–	–
		WinCLIP (reported)	0.838	–	–	–
		Controlled DINOv2 NN	0.804	0.809	0.434	2.005
		CRR / PCA64	0.823	0.834	0.429	0.472
		CRR / PCA128	0.834	0.842	0.430	0.566
	4	Official AnomalyDINO	0.912	0.918	–	–
		WinCLIP (reported)	0.873	–	–	–
		Controlled DINOv2 NN	0.862	0.861	0.543	6.000
		CRR / PCA64	0.870	0.883	0.284	0.472
		CRR / PCA128	0.885	0.895	0.278	0.566
	8	Official AnomalyDINO	0.926	0.929	–	–
		Controlled DINOv2 NN	0.873	0.870	0.533	6.000
		CRR / PCA64	0.880	0.891	0.205	0.472
		CRR / PCA128	0.897	0.905	0.170	0.566

ranking improvement unless the raw score changes. Conversely, ranking methods cannot claim calibrated probabilities unless ECE/Brier/NLL improve. This separation prevents a common ambiguity in AD papers: conflating a better sorted list with a more trustworthy alarm probability.

5. Results

5.1. Clean Accuracy and Storage

Table 1 summarizes clean image-level performance and storage. The official AnomalyDINO reproduction (released code, 3 seeds) leads image AUROC at every k on both datasets (MVTec 0.965/0.976/0.980; VisA 0.857/0.912/0.926, matching its published VisA numbers), which we report as an explicit accuracy reference: neither the controlled nearest-neighbor baseline nor CRR claims ranking superiority. CalibSubspaceHead is competitive with the controlled DINOv2 nearest-neighbor memory-bank baseline while being far more compact: PCA64 stores about 0.472 MB, compared with roughly 2–6 MB for the controlled memory bank. On VisA, the detector is stronger relative to this controlled baseline, and the PCA128 ablation improves VisA AUROC by 0.011–0.016 over PCA64 while keeping storage below 0.6 MB; a paired class–seed hierarchical bootstrap confirms this gain is statistically supported, with 95% intervals excluding zero at every k ($k=1$: [+0.003, +0.020]; $k=2$: [+0.006, +0.026]; $k=4$: [+0.009, +0.024]; $k=8$: [+0.010, +0.019]). The controlled rows must not be read as official PatchCore or AnomalyDINO reproductions; only the official AnomalyDINO rows are. Two entries in Table 1 deserve explicit anatomy so they are not mistaken for placeholders or a strawman. Under the label-free scalar-Platt protocol, the controlled memory-bank baseline’s probability collapses to a constant: at $k \leq 2$ every support patch self-matches (distance zero, probability ≈ 1), while at $k \geq 4$ the 4096-patch memory cap forces subsampling (probability ≈ 0). Its ECE therefore equals a prevalence complement (0.277/0.693) with Brier $\approx$

Table 2

Full VisA conformal reliability under corruptions. The PCA/subspace ranking score is fixed; only the reliability view changes. Best reliability values within each group are bold.

Group	Reliability view	N	AUROC	AP	ECE ↓	Brier ↓	NLL ↓
All	LOIO conformal	56000	0.823	0.846	0.077	0.187	0.686
All	Weighted conformal	56000	0.823	0.846	0.166	0.244	0.813
k=4	LOIO conformal	28000	0.820	0.842	0.039	0.187	0.760
k=4	Weighted conformal	28000	0.820	0.842	0.216	0.259	0.925
k=8	LOIO conformal	28000	0.828	0.851	0.114	0.188	0.612
k=8	Weighted conformal	28000	0.828	0.851	0.116	0.229	0.701

ECE—the signature of a constant probability—and its storage saturates at exactly 6.000 MB (4096 patches $\times$ 384 dims $\times$ 4 bytes). These values diagnose that raw nearest-neighbor distances resist naive label-free calibration; they are not competitive calibration results.

Storage accounting and compute cost. The per-class figures above cover only the class-specific reference state and deliberately exclude the shared frozen DINOv2 ViT-S/14 backbone (≈ 22 M parameters, ≈ 84 MB in fp32), which every compared method requires once per deployment. The 0.472 MB CRR state decomposes into the PCA subspace (64 components $\times$ 384 dims in fp32 ≈ 0.098 MB, matching the observed 0.094 MB PCA128–PCA64 increment) and the synthetic-anomaly head and calibrator (≈ 0.37 MB); the controlled memory bank stores up to 4096 patch vectors (≤ 6.0 MB). At inference, cached-feature scoring costs about 1.3 ms per image for the subspace detector versus 5–13 ms for the nearest-neighbor bank, while the shared backbone forward dominates end-to-end latency (order of seconds per batch on first pass, identical across methods). Fitting the k leave-one-out subspaces for LOIO adds a one-time per-class setup cost of k PCA fits on at most $k \times 1369$ patch vectors, i.e., well under a second at $k \leq 8$.

Aggregation sensitivity. Because the clean benchmark aggregates patch residuals by max while the conformal pipeline uses the top-1% mean, Table 13 (Appendix A) quantifies the sensitivity of the raw ranking to this choice on representative classes. The spread across aggregators is 0.01–0.02 AUROC: moderate top- ρ pooling ($\rho \in [0.005, 0.02]$) is slightly better than max on MVTec, while very wide pooling ($\rho=0.05$) hurts VisA. Both configurations used in the paper sit within 0.01–0.02 of the per-setting optimum, so no conclusion in this paper depends on the aggregator choice.

5.2. Full VisA Conformal Reliability

Table 2 gives the main calibration-error result. Throughout the paper, ECE uses 15 equal-width bins; because few-shot conformal confidences occupy at most $k+1$ distinct values, we also report an equal-mass adaptive ECE below, and Section 5.4 replaces ECE with conformal-native diagnostics for the central claim. Across 56,000 full VisA corruption rows, LOIO conformal obtains pooled ECE 0.0766 while preserving the same PCA/subspace ranking. The strongest regime is $k=4$, where pooled ECE drops to 0.0391. At $k=8$, ECE increases to 0.1140 but remains better than Platt-style baselines in every corruption cell. The pooled reliability diagrams in Figure 2 visualize both effects: confidences occupy only the $k+1$ attainable levels, and the gap below the diagonal—overconfidence on normal images—widens from $k=4$ to $k=8$ and is larger on MVTec than on VisA. The uniformity analysis in Section 5.4 explains this k trend mechanistically: with a clean support and corrupted test images, normal test residuals are inflated relative to the clean LOIO calibration set; the tighter $k=8$ subspace amplifies this mismatch, pushing normal p-values anti-conservative.

Table 12 (Appendix A) isolates corruption effects. LOIO conformal improves over Vector Platt and Shift-Aware Platt for blur, brightness/contrast shift, Gaussian noise, and JPEG at both $k=4$ and $k=8$. Weighted conformal is a useful ablation but not a safe default: it is competitive for selected $k=8$ structured corruptions, yet it is much worse at $k=4$ and under Gaussian noise.

5.3. Full MVTec Conformal Reliability

The same conformal protocol was run on all 15 MVTec classes ($k \in \{4, 8\}$, seeds 0–2, four corruptions, 37,464 image rows). The VisA conclusion replicates: LOIO conformal reliability attains an overall ECE of 0.0684 (0.0600 at $k=4$, 0.0769 at $k=8$) while preserving the raw ranking (image AUROC 0.912), and it improves ECE over both Vector Platt and Shift-Aware Platt in every one of the eight k –corruption cells, often by a factor of two to four (e.g., $k=4$ Gaussian noise: 0.033 versus 0.233 and 0.270; Figure 3). Weighted conformal is again markedly worse (ECE 0.31–0.45). Conformal reliability is therefore not a VisA-specific effect; it holds across both benchmarks under identical protocols.

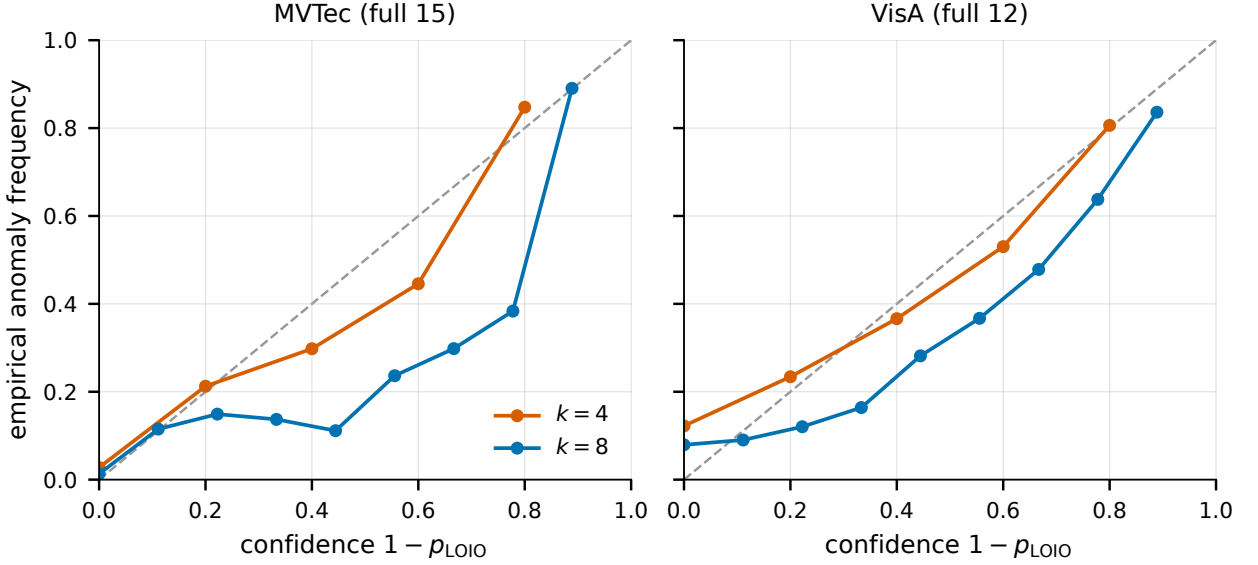


Figure 2: Reliability diagrams for LOIO conformal reliability on the full MVTec (left) and VisA (right) corruption benchmarks, pooled over classes, seeds, and corruptions. Points occupy the $k+1$ attainable confidence levels; gaps below the diagonal indicate overconfidence, which is larger at $k=8$, consistent with the uniformity analysis in Section 5.4.

5.4. Statistical Validity: Uniformity Tests, Paired Significance, and Standard Calibrators

Conformal p-values promise $P(p \leq \alpha) \leq \alpha$ on normal data under exchangeability—a false-alarm guarantee, not a posterior probability. We therefore test this property directly instead of relying on ECE. Because a k -shot LOIO p-value is discrete on the grid $\{j/(k+1)\}$, a continuous Kolmogorov–Smirnov test against $U(0, 1)$ is invalid by construction; we instead compare the empirical CDF of normal p-values to its exact expectation at every grid point and calibrate the resulting discrete KS statistic against the exact null by Monte Carlo. Figure 4 shows the resulting discrete Q-Q view. Uniformity is rejected in all 16 dataset– k –corruption cells (Monte Carlo $p \leq 0.005$), but the *direction* of the violation is the informative result. On VisA at $k=4$ the deviation is uniformly conservative (empirical CDF below nominal; worst gap -0.10 at the 0.4 grid point), consistent with the below-nominal false-alarm rates above. On VisA at $k=8$ and on MVTec at both k , the deviation flips anti-conservative under corruption—most severely for Gaussian noise on MVTec $k=4$, where $\hat{F}(0.2) = 0.46$ against a nominal 0.20. This is exactly the clean-support/corrupted-test mechanism anticipated by the exchangeability caveat in the method section: corruption inflates normal test residuals relative to the clean LOIO calibration set, and a tighter $k=8$ subspace amplifies the mismatch. A matched-LOIO audit that scores support and test under the same fold-specific subspaces (the jackknife+ pairing [31]) does not repair this trade-off: on representative MVTec $k=4$ it raises clean false alarms from 0.17 to 0.25 with essentially unchanged power. The violation is driven by the support–test condition mismatch, not by the leave-one-out asymmetry alone—which is precisely the failure mode SC3R’s matched-condition source validation addresses.

Paired significance replaces point estimates for the headline calibration comparison, and the comparison now covers the full standard toolbox: Table 3 reports class–seed–corruption cell ECE for LOIO against Vector Platt, Shift-Aware Platt, and the standard scalar calibrators—temperature scaling, isotonic regression, and histogram binning—all fit on the identical label-free calibration set. LOIO conformal is significantly better than every scalar calibrator on both benchmarks and both k (paired Wilcoxon $p \leq 2.5 \times 10^{-4}$ in all 16 comparisons, and $p \leq 4.8 \times 10^{-8}$ in 14 of them; e.g., MVTec $k=4$: LOIO 0.120 ± 0.054 versus isotonic 0.222 ± 0.088 , temperature 0.240 ± 0.104 , histogram binning 0.229 ± 0.099). Among the scalar family, isotonic regression is strongest and narrows the gap at $k=8$ (MVTec: delta -0.023 , LOIO better in only 55% of cells, though still $p=2.5 \times 10^{-4}$; VisA: delta -0.035 , 62%), which is expected: with $2k$ calibration points the nonparametric fit becomes more competitive as k grows. Against Vector Platt, LOIO wins with $p \leq 4.9 \times 10^{-9}$ everywhere; against Shift-Aware Platt the result holds everywhere except VisA $k=8$, where the difference is not significant ($p=0.20$, LOIO better in 51% of cells)—an honest boundary we state rather than average away. Replacing equal-width with equal-mass (adaptive) ECE, which cannot be flattered by at most $k+1$ occupied

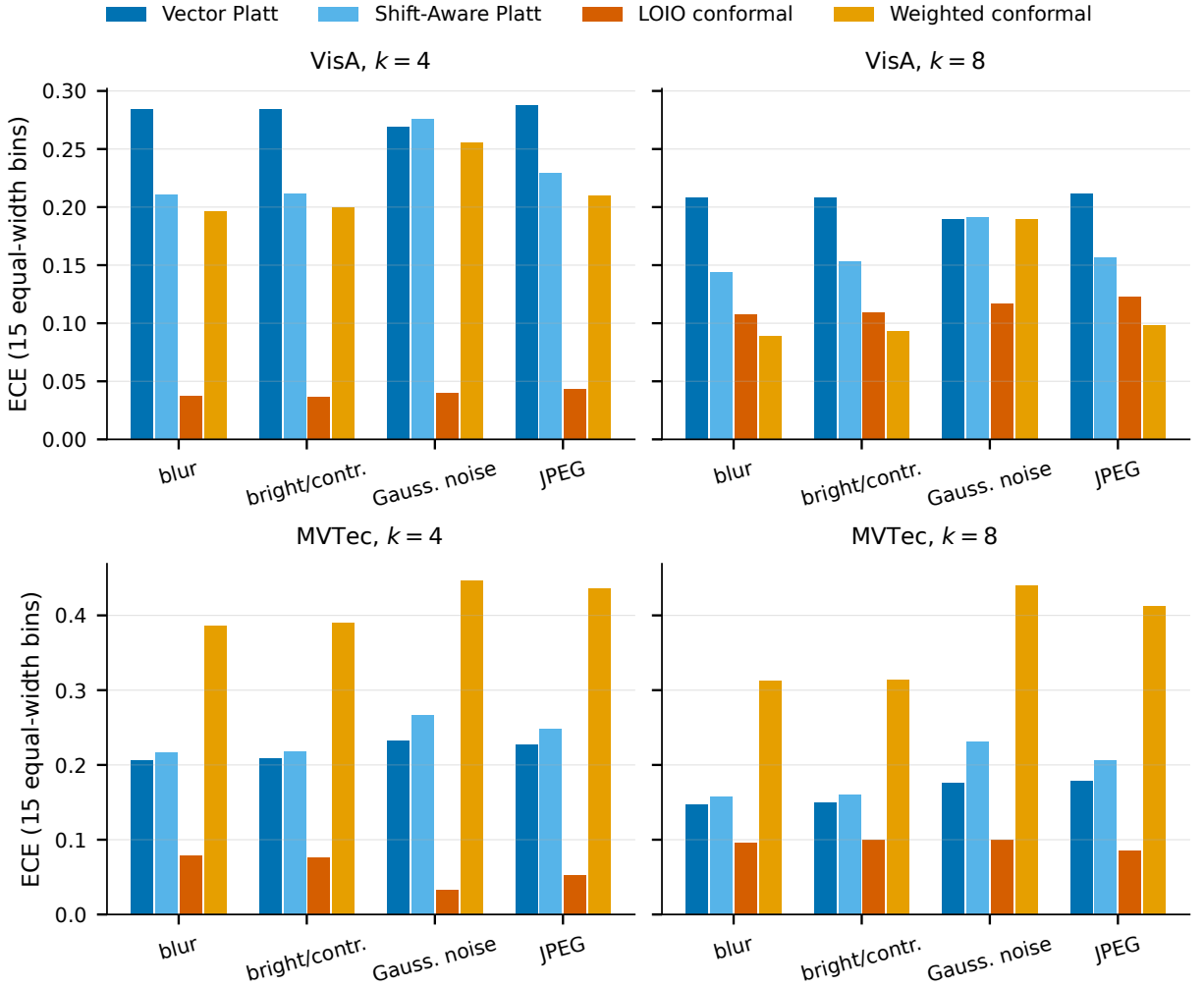


Figure 3: ECE by corruption and reliability view on full VisA (top) and full MVTEC (bottom) at $k=4$ and $k=8$. LOIO conformal is the most stable view; weighted conformal is competitive only in selected VisA $k=8$ structured corruptions.

confidence levels, increases LOIO’s per-cell ECE (MVTEC: $0.120 \rightarrow 0.160$ at $k=4$, $0.135 \rightarrow 0.182$ at $k=8$; VisA: $0.132 \rightarrow 0.227$, $0.158 \rightarrow 0.236$) while leaving the ordering against all calibrator baselines intact. Binning does inflate the apparent margin, which is one more reason the operational analyses below, not ECE, carry the central claim.

5.5. Attainable Alpha and Operational False-Alarm Control

Calibration error alone does not answer the operator’s question: “if I alarm at $p \leq \alpha$, how often is a good part flagged?” Table 4 reports the attainable-alpha analysis. The pattern is structural, not empirical: at $k=4$ every nominal level below 0.2 raises zero alarms on both datasets because no attainable p-value exists below the floor $1/(k+1)$, and at $k=8$ the same holds below $1/9$. Reporting these zeros as “conservative coverage” would be misleading; they are resolution artifacts of the few-shot calibration set.

Table 5 evaluates the first attainable operating point, $\alpha=0.20$, on both full benchmarks. On VisA the LOIO alarm rule is conservative but functional (VisA $k=4$: false-alarm 0.14–0.16 with detection 0.58–0.61 and precision around 0.81); on full MVTEC the same rule is anti-conservative under corruption at $k=4$, which the SC3R analysis below addresses directly. We note a practical pitfall uncovered during this analysis: p-values stored in single precision can exceed their exact rational value (e.g., $1/5 \rightarrow 0.20000000298$), so an exact comparison $p \leq \alpha$ silently discards every alarm at the floor. All thresholding in this paper uses a small tolerance, and we recommend the same in deployments.

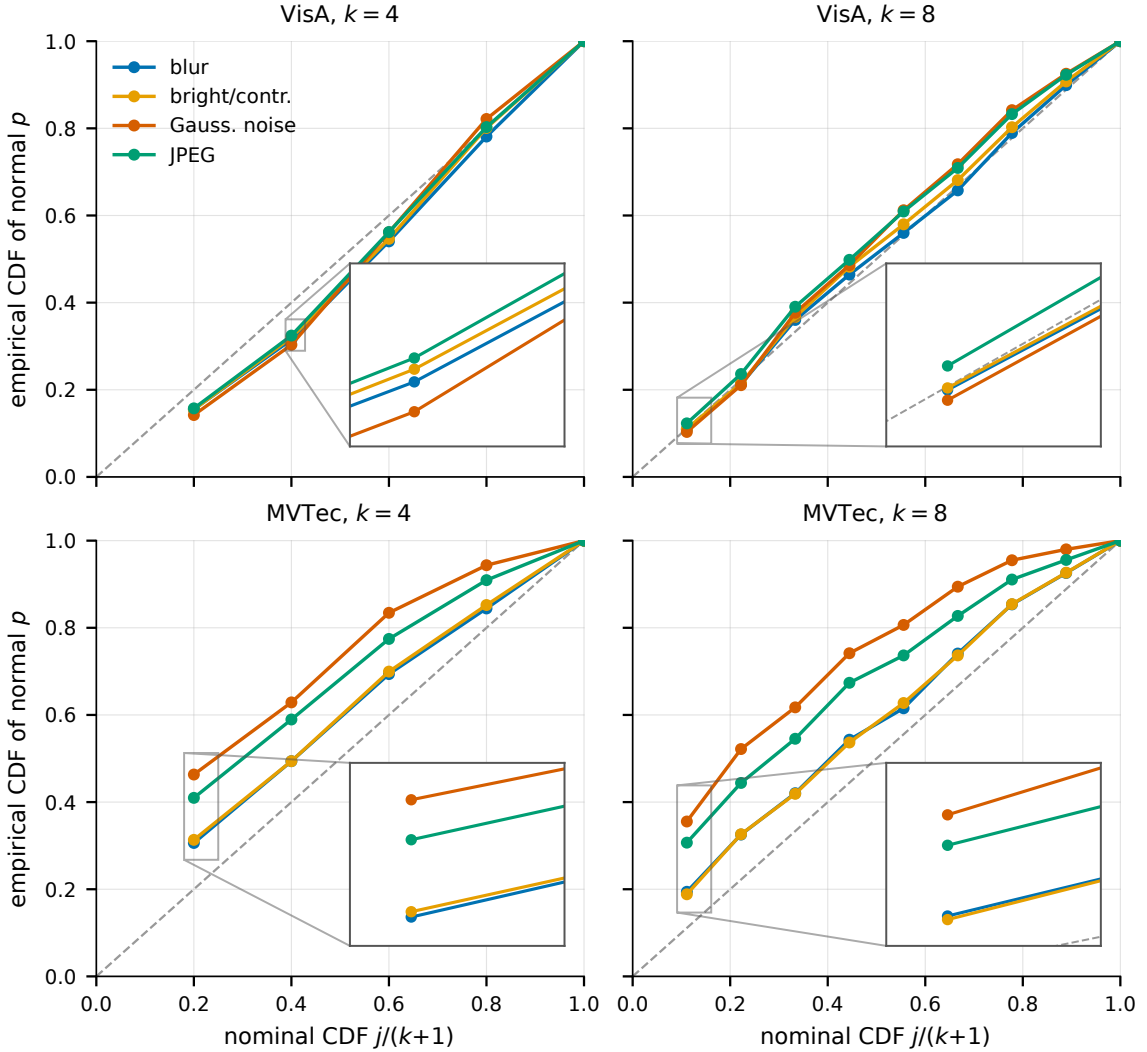


Figure 4: Discrete Q-Q analysis of normal-image LOIO p-values: empirical CDF at each attainable grid point $j/(k+1)$ against its exact null expectation (dashed diagonal), per corruption. Points above the diagonal are anti-conservative (excess false alarms). VisA $k=4$ is conservative; VisA $k=8$ and MVTec drift anti-conservative under corruption, worst for Gaussian noise. Insets give a tight zoom around the first attainable grid point $1/(k+1)$, where the curves overlap most.

5.6. SC3R: Operating Below the Target-Only Alpha Floor

Table 6 evaluates SC3R against the target-only LOIO anchor on all 15 label-stratified MVTec classes ($k=4$, seeds 0–2, five conditions, 675 evaluation cells), under a pre-registered decision gate fixed before the run: nonzero power at $\alpha \in \{0.05, 0.10\}$, mean false-alarm rate at most $\alpha+0.02$, no-harm versus the anchor in at least 80% of class–seed–corruption cells, no target anomaly labels, and hierarchical class–seed bootstrap support. Matched-condition SC3R passes every criterion. It achieves mean power 0.216 at $\alpha=0.05$ and 0.416 at $\alpha=0.10$ in a regime where the target-only anchor is structurally silent, while mean false-alarm rates track nominal almost exactly: 0.050, 0.105, and 0.216 at $\alpha \in \{0.05, 0.10, 0.20\}$, with per-condition false-alarm rates between 0.048 and 0.051 at $\alpha=0.05$ and alarm precision above 0.90 throughout. No-harm rates are 89%, 82%, and 89%. The hierarchical class–seed bootstrap is decisive at this scale: the 95% confidence interval for the power gain over the anchor excludes zero for *every* condition at both sub-floor levels (e.g., clean $\alpha=0.10$: [0.30, 0.64]; JPEG $\alpha=0.05$: [0.02, 0.28]). At $\alpha=0.20$, at the anchor’s floor, the roles reverse in an instructive way: the anchor attains higher power but its false-alarm rate inflates to 0.27–0.46 under corruption, well above nominal, whereas SC3R remains within 0.005–0.035 of nominal. Source validation thus buys consistent

Table 3

LOIO conformal reliability versus Platt-family and standard scalar calibrators on the full corruption benchmarks. All calibrators are label-free and fit on the identical calibration set (k support normals plus k synthetic anomalies). Entries are means $\pm$ std of per-cell (class $\times$ seed $\times$ corruption) ECE with 15 equal-width bins—per-cell means are therefore higher than the pooled ECE of Table 2; Δ is LOIO minus baseline (negative favors LOIO), p is a paired two-sided Wilcoxon signed-rank test, and the last column is the fraction of cells where LOIO is better. Bold marks the better per-cell mean; the only comparison that is not significant is Shift-Aware Platt at VisA $k=8$.

dataset / k	baseline	base ECE	LOIO ECE	Δ	Wilcoxon p	LOIO better
MVTec $k=4$	Vector Platt	0.223 $\pm$ 0.095	0.120$\pm$0.054	-0.104	2×10^{-23}	84%
	Shift-Aware Platt	0.241 $\pm$ 0.104	0.120$\pm$0.054	-0.121	9×10^{-25}	88%
	Temperature scaling	0.240 $\pm$ 0.104	0.120$\pm$0.054	-0.120	1×10^{-25}	87%
	Isotonic regression	0.222 $\pm$ 0.088	0.120$\pm$0.054	-0.103	1×10^{-24}	87%
	Histogram binning	0.229 $\pm$ 0.099	0.120$\pm$0.054	-0.110	1×10^{-23}	84%
	Scalar Platt	0.268 $\pm$ 0.127	0.120$\pm$0.054	-0.148	3×10^{-29}	93%
MVTec $k=8$	Vector Platt	0.167 $\pm$ 0.065	0.135$\pm$0.055	-0.033	5×10^{-9}	66%
	Shift-Aware Platt	0.192 $\pm$ 0.088	0.135$\pm$0.055	-0.057	9×10^{-15}	75%
	Temperature scaling	0.177 $\pm$ 0.080	0.135$\pm$0.055	-0.042	5×10^{-10}	69%
	Isotonic regression	0.157 $\pm$ 0.069	0.135$\pm$0.055	-0.023	2×10^{-4}	55%
	Histogram binning	0.162 $\pm$ 0.080	0.135$\pm$0.055	-0.027	4×10^{-5}	61%
	Scalar Platt	0.256 $\pm$ 0.132	0.135$\pm$0.055	-0.121	2×10^{-27}	91%
VisA $k=4$	Vector Platt	0.286 $\pm$ 0.060	0.131$\pm$0.063	-0.155	6×10^{-24}	97%
	Shift-Aware Platt	0.237 $\pm$ 0.087	0.131$\pm$0.063	-0.106	3×10^{-19}	90%
	Temperature scaling	0.312 $\pm$ 0.085	0.132$\pm$0.059	-0.180	1×10^{-40}	98%
	Isotonic regression	0.239 $\pm$ 0.073	0.132$\pm$0.059	-0.106	4×10^{-33}	90%
	Histogram binning	0.301 $\pm$ 0.087	0.132$\pm$0.059	-0.168	4×10^{-40}	96%
	Scalar Platt	0.304 $\pm$ 0.145	0.132$\pm$0.059	-0.171	4×10^{-35}	90%
VisA $k=8$	Vector Platt	0.199 $\pm$ 0.063	0.151$\pm$0.054	-0.048	3×10^{-14}	81%
	Shift-Aware Platt	0.162 $\pm$ 0.075	0.151$\pm$0.054	-0.011	0.20	51%
	Temperature scaling	0.247 $\pm$ 0.089	0.158$\pm$0.051	-0.089	7×10^{-23}	75%
	Isotonic regression	0.194 $\pm$ 0.068	0.158$\pm$0.051	-0.035	5×10^{-8}	62%
	Histogram binning	0.242 $\pm$ 0.097	0.158$\pm$0.051	-0.083	2×10^{-18}	69%
	Scalar Platt	0.418 $\pm$ 0.100	0.158$\pm$0.051	-0.259	1×10^{-40}	98%

false-alarm tracking at every level, at the cost of some power at levels the anchor can reach. Residual weaknesses are small and localized: JPEG at $\alpha=0.10$ (0.121) and Gaussian noise/JPEG at $\alpha=0.20$ (0.226/0.235) exceed the $\alpha+0.02$ budget per-condition while the pooled means stay within it.

Replication at $k=8$. Table 7 repeats the protocol at $k=8$ (floor $1/9 \approx 0.111$), where the anchor is silent at $\alpha \in \{0.05, 0.10\}$. The sub-floor power unlocking replicates and strengthens: matched-condition SC3R reaches mean power 0.547 at $\alpha=0.05$ and 0.693 at $\alpha=0.10$ (clean/blur/brightness 0.75–0.83) with precision above 0.91, and the hierarchical bootstrap interval for the power gain again excludes zero for every corruption at both sub-floor levels (weakest: Gaussian noise $\alpha=0.05$, [0.05, 0.29]). False-alarm tracking is exact on clean and structured corruptions (0.050–0.053 at $\alpha=0.05$) but the pre-registered budget is met only partially: the pooled mean passes at $\alpha=0.05$ (0.052) yet misses the $\alpha+0.02$ budget marginally at $\alpha=0.10$ (0.121) and $\alpha=0.20$ (0.230), with no-harm rates 80%/74%/81%, and the exceedances again concentrate in Gaussian noise and JPEG (0.138/0.141 at $\alpha=0.10$). At $\alpha=0.20$ the anchor over-alarms under exactly these corruptions (0.39/0.38) while SC3R stays at 0.23–0.26. The mechanism behind the residual exceedances is the same one the uniformity tests expose for the anchor, attenuated but not eliminated: Gaussian noise and JPEG inflate normal residuals more heterogeneously across classes than blur or brightness shifts, so a threshold validated on held-out source classes transfers slightly optimistically to targets whose noise response sits in the tail—and the tighter $k=8$ subspace widens exactly this spread. Per-condition (rather than pooled) source validation is the natural mitigation and we flag it as future work. We therefore state the SC3R claim with its boundary: the gate passes in full at $k=4$ and at $k=8$ for $\alpha=0.05$, while heavy-tailed corruptions erode the false-alarm budget at higher levels—the same two conditions flagged at $k=4$.

Why source validation rather than randomization. The standard remedy for the attainable-alpha floor is the smoothed conformal p-value, which breaks the $k+1$ -point discreteness with an independent uniform tie-break and admits alarms at any nominal level. Table 8 evaluates this baseline on the same target-only LOIO residuals and test

Table 4

Attainable-alpha structure of few-shot LOIO conformal p-values. With k support scores the smallest attainable p-value is $1/(k+1)$, so nominal levels below this floor structurally raise no alarms; empirical false-alarm rates (FAR, mean over classes/seeds/corruptions) then sit at the nearest attainable grid point rather than at the nominal level.

dataset	k	nominal	α	floor	attainable	FAR	detection
VisA (full)	4	0.01	0.200	0.000	0.000	0.000	0.000
	4	0.05	0.200	0.000	0.000	0.000	0.000
	4	0.10	0.200	0.000	0.000	0.000	0.000
	4	0.20	0.200	0.200	0.152	0.599	0.599
	8	0.01	0.111	0.000	0.000	0.000	0.000
	8	0.05	0.111	0.000	0.000	0.000	0.000
	8	0.10	0.111	0.000	0.000	0.000	0.000
	8	0.20	0.111	0.111	0.111	0.537	0.537
MVTec (full)	4	0.01	0.200	0.000	0.000	0.000	0.000
	4	0.05	0.200	0.000	0.000	0.000	0.000
	4	0.10	0.200	0.000	0.000	0.000	0.000
	4	0.20	0.200	0.200	0.411	0.887	0.887
	8	0.01	0.111	0.000	0.000	0.000	0.000
	8	0.05	0.111	0.000	0.000	0.000	0.000
	8	0.10	0.111	0.000	0.000	0.000	0.000
	8	0.20	0.111	0.111	0.304	0.907	0.907

Table 5

Operational false-alarm behavior of target-only LOIO conformal p-values at the first attainable operating point ($\alpha=0.20$; alarms fire at $p \leq \alpha$ with a float tolerance). Rates are pooled over classes and seeds. On VisA the rule is conservative (FAR below nominal); on full MVTec it is anti-conservative under corruption, especially at $k=4$, motivating source-validated thresholding (Table 6).

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.155	0.610	0.806
	blur	8	0.109	0.546	0.841
	bright/contr.	4	0.155	0.595	0.803
	bright/contr.	8	0.110	0.533	0.836
	Gauss. noise	4	0.142	0.585	0.813
	Gauss. noise	8	0.103	0.522	0.843
	JPEG	4	0.157	0.604	0.803
	JPEG	8	0.123	0.547	0.825
MVTec (full)	blur	4	0.306	0.887	0.872
	blur	8	0.194	0.908	0.916
	bright/contr.	4	0.313	0.887	0.869
	bright/contr.	8	0.188	0.908	0.919
	Gauss. noise	4	0.463	0.879	0.816
	Gauss. noise	8	0.355	0.898	0.855
	JPEG	4	0.410	0.890	0.836
	JPEG	8	0.307	0.914	0.875

images as SC3R, reporting exact expectations over the randomization. Randomization does cross the floor, and on VisA—where the target-only rule is conservative—it behaves reasonably (FAR below nominal at reduced power). But it has no mechanism to correct the support–test mismatch: on MVTec its false-alarm rate exceeds the budget *even under the clean condition* (0.067 at $\alpha=0.05$, reflecting the leave-one-out asymmetry) and reaches 2.3× nominal under Gaussian noise (0.115), inheriting exactly the anti-conservative bias that the uniformity tests localize. SC3R attains comparable sub-floor power (0.216 vs. 0.216 at $\alpha=0.05$; 0.416 vs. 0.431 at $\alpha=0.10$) while keeping false alarms at nominal on both datasets. Randomization solves the resolution problem; source validation solves the resolution *and* the validity problem.

Replication on VisA and cross-dataset source archives. Table 9 repeats the full protocol on all 12 label-stratified VisA classes ($k=4$, seeds 0–2), with two source pools. With within-VisA sources, the MVTec conclusion replicates: false-alarm rates track nominal closely (0.051/0.095/0.192—at and below nominal at every level, unlike MVTec’s

Table 6

Source-validated thresholding (SC3R, matched-condition source mode) versus the target-only LOIO anchor on all 15 label-stratified MVTec classes ($k=4$, seeds 0–2). Below the target-only attainable-alpha floor $1/(k+1)=0.2$, the anchor cannot raise alarms; source pooling unlocks these operating points with controlled false alarms. Bold marks cells satisfying $\text{FAR} \leq \alpha + 0.02$ with nonzero power.

corruption	α	SC3R (source-validated)			Target-only anchor		
		FAR	power	prec.	FAR	power	prec.
clean	0.05	0.050	0.267	0.928	0.000	0.000	–
	0.10	0.100	0.470	0.957	0.000	0.000	–
	0.20	0.209	0.768	0.914	0.267	0.863	0.898
blur	0.05	0.048	0.242	0.915	0.000	0.000	–
	0.10	0.099	0.468	0.950	0.000	0.000	–
	0.20	0.207	0.760	0.904	0.269	0.860	0.895
bright/contr.	0.05	0.050	0.254	0.914	0.000	0.000	–
	0.10	0.101	0.467	0.954	0.000	0.000	–
	0.20	0.205	0.769	0.905	0.268	0.863	0.898
Gauss. noise	0.05	0.048	0.184	0.934	0.000	0.000	–
	0.10	0.105	0.295	0.929	0.000	0.000	–
	0.20	0.226	0.495	0.914	0.460	0.860	0.844
JPEG	0.05	0.051	0.134	0.911	0.000	0.000	–
	0.10	0.121	0.381	0.936	0.000	0.000	–
	0.20	0.235	0.633	0.917	0.434	0.867	0.854

Table 7

SC3R at $k=8$ (matched-condition source mode) versus the target-only LOIO anchor on all 15 label-stratified MVTec classes (seeds 0–2). The target-only floor is $1/9 \approx 0.111$, so $\alpha \in \{0.05, 0.10\}$ are below the anchor’s resolution. Bold marks cells satisfying $\text{FAR} \leq \alpha + 0.02$ with nonzero power.

corruption	α	SC3R (source-validated)			Target-only anchor		
		FAR	power	prec.	FAR	power	prec.
clean	0.05	0.053	0.764	0.976	0.000	0.000	–
	0.10	0.110	0.833	0.936	0.000	0.000	–
	0.20	0.216	0.898	0.917	0.182	0.903	0.928
blur	0.05	0.050	0.751	0.976	0.000	0.000	–
	0.10	0.110	0.813	0.957	0.000	0.000	–
	0.20	0.221	0.900	0.917	0.199	0.904	0.922
bright/contr.	0.05	0.051	0.770	0.976	0.000	0.000	–
	0.10	0.107	0.831	0.958	0.000	0.000	–
	0.20	0.223	0.895	0.916	0.185	0.904	0.926
Gauss. noise	0.05	0.035	0.155	0.962	0.000	0.000	–
	0.10	0.138	0.398	0.951	0.000	0.000	–
	0.20	0.232	0.749	0.917	0.392	0.904	0.866
JPEG	0.05	0.071	0.296	0.972	0.000	0.000	–
	0.10	0.141	0.588	0.959	0.000	0.000	–
	0.20	0.260	0.826	0.918	0.380	0.911	0.875

Gaussian/JPEG exceedances), sub-floor power is 0.144/0.351 with alarm precision above 0.84, no-harm is 89%/84% at the sub-floor levels, and the hierarchical power-gain interval again excludes zero for every corruption at both sub-floor levels (e.g., blur $\alpha=0.10$: [0.21, 0.50]). At $\alpha=0.20$ no-harm dips to 78%, but the failure mode differs from MVTec’s: the VisA anchor is itself conservative at its floor, so SC3R occasionally alarms slightly more than an anchor that under-alarms, while remaining below nominal. Replacing the VisA source pool with the 15 MVTec classes—a genuine cross-dataset source archive—answers the transfer question directly: the archive is *conservative*, with false-alarm rates well below nominal (0.013/0.040/0.098), roughly half to two-thirds of the within-dataset power (0.091/0.201/0.415), precision above 0.85, no-harm 93–97%, and power-gain intervals that still exclude zero for every corruption at both sub-floor levels (weakest: JPEG $\alpha=0.05$, [0.005, 0.066]). Cross-dataset source archives therefore fail safe: when only

Table 8

Why source validation, not randomization: smoothed (randomized) conformal p-values versus SC3R below the target-only floor ($k=4$, all classes, seeds 0–2, pooled over five conditions; randomized rates are exact expectations over the randomization). Randomization crosses the floor but inherits the target-only rule's bias—anti-conservative on MVTec even under the clean condition (FAR 0.067 at $\alpha=0.05$) and up to 2.3 $\times$ nominal under Gaussian noise—whereas SC3R attains comparable power with false-alarm rates at nominal. Bold marks FAR $\leq \alpha+0.02$.

dataset	α	Randomized p-value		SC3R	
		FAR	power	FAR	power
MVTec	0.05	0.085	0.216	0.049	0.216
	0.10	0.170	0.431	0.105	0.416
VisA	0.05	0.036	0.150	0.051	0.144
	0.10	0.072	0.300	0.095	0.351

Table 9

SC3R replication on all 12 label-stratified VisA classes ($k=4$, seeds 0–2, matched-condition mode, pooled over five conditions). The source pool is either the other VisA classes (within-dataset) or all 15 MVTec classes (cross-dataset source archive). Both sub-floor levels carry hierarchical power-gain intervals excluding zero for every corruption; the cross-dataset archive is conservative, keeping false alarms well below nominal at reduced power. Bold marks FAR $\leq \alpha+0.02$ with nonzero power.

source pool	α	FAR	power	prec.	no-harm
VisA (within)	0.05	0.051	0.144	0.847	89%
	0.10	0.095	0.351	0.843	84%
	0.20	0.192	0.637	0.810	78%
MVTec (cross-dataset)	0.05	0.013	0.091	0.936	97%
	0.10	0.040	0.201	0.881	93%
	0.20	0.098	0.415	0.852	93%

foreign-category normals are available, SC3R trades power for a false-alarm rate that stays under budget rather than over it.

5.7. ECE Is Prevalence-Sensitive for Conformal Scores

A further protocol warning emerged from prevalence stress tests: ECE computed on $1-p$ conformal scores changes drastically with the anomaly prevalence of the evaluation set even when the ranking is fixed. On full VisA, LOIO ECE moves from 0.404 at 1% prevalence to 0.149 at 50%, and the LOIO-versus-weighted ranking reverses across this range. ECE on anomaly-balanced test sets therefore cannot certify deployment reliability of conformal-derived scores. We report ECE as a secondary metric only and rely on false-alarm rate, power, precision, and risk-coverage for operational claims. The risk-coverage view (Figure 5) also quantifies the value of abstention: ranking images by predictive entropy and abstaining on the most uncertain 20% halves the remaining LOIO calibration error on full MVTec (0.068 to 0.030) and improves it moderately on VisA (0.077 to 0.075 at 80% coverage, 0.066 at 70%), so selective operation is a practical complement to fixed- α alarms.

5.8. Routing, Pixel Localization, and Fragility Diagnostics

Table 10 evaluates protocol-locked representative routing. Fixed LOIO conformal reduces ECE relative to Vector Platt without using validation labels to select an expert. This supports the central CRR claim: reliability routing can be useful even when the route is fixed and conservative. SAGE-style gating remains a promising design inspiration, but the current main evidence favors simple LOIO routing over learned or weighted expert selection.

Pixel-level results (Table 11) show that the low-storage subspace detector is also competitive for localization: it matches or exceeds the controlled memory bank on pixel AUROC at every k and on AU-PRO from $k=4$. Robustness diagnostics are less favorable: the label-aware PCA-residual FGSM surrogate collapses image AUROC from 0.94 to 0.16 at $\epsilon=2/255$ —far below chance, i.e., the attack inverts the score ordering—and the drop shrinks rather than grows at larger ϵ (0.45 at 8/255), a non-monotonicity that signals objective sensitivity of the surrogate itself. These results

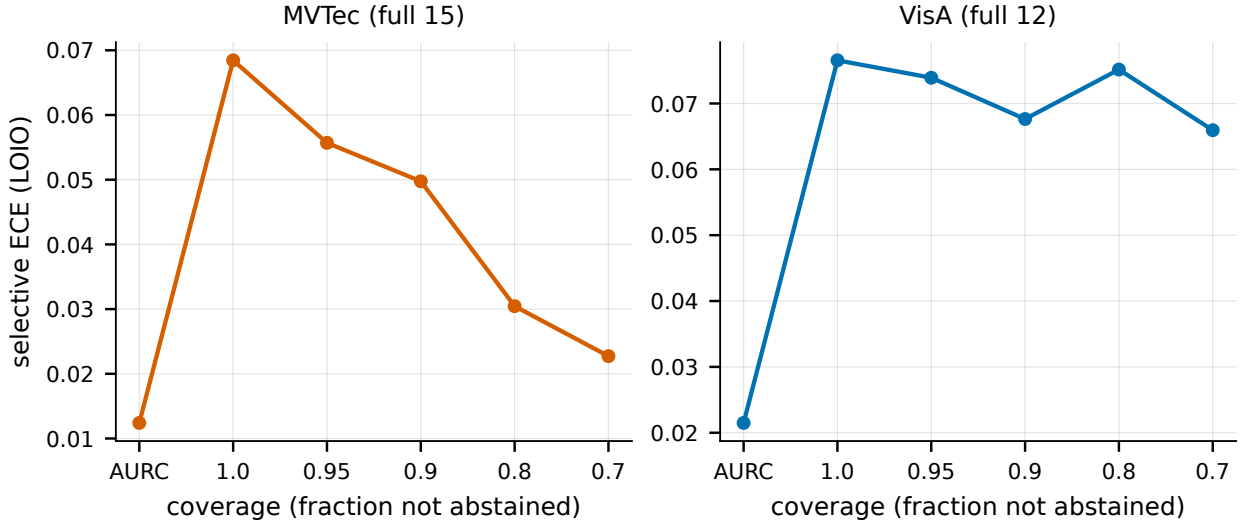


Figure 5: Risk-coverage curves for LOIO conformal reliability with entropy-ranked abstention on full MVTEC (left) and full VisA (right). Abstaining on the 20% most uncertain images halves the remaining calibration error on MVTEC.

Table 10

Protocol-locked representative routing. Fixed LOIO conformal does not require validation-label expert selection and reduces ECE relative to Vector Platt in every protocol; no-harm is the fraction of cells where the LOIO route does not degrade calibration.

Protocol	Vector ECE	LOIO ECE	Δ ECE	No-harm
MVTEC→VisA	0.301	0.099	-0.203	100%
VisA→MVTEC	0.304	0.102	-0.202	100%
LOCO	0.304	0.116	-0.188	100%
Within split	0.361	0.154	-0.207	100%

are important negative evidence and motivate reliability diagnostics, but they do not support an adversarial robustness claim.

5.9. Novelty Guardrails

Table 14 (Appendix A) marks, for each ingredient, where prior work ends and this paper’s claims begin. The official SubspaceAD representative run is especially important because it reaches high image and pixel AUROC in our protocol: the ranking mechanism is inherited, and the contributions of this paper start after it—at the attainable-alpha floor, its uniformity audit, and the source-validated thresholding that crosses it.

6. Limitations

CRR is intended as a reliability framework, not as a universal accuracy replacement. First, the method is not an MVTEC AUROC SOTA claim. Official SubspaceAD remains a very strong ranking baseline, and our MVTEC contribution is mainly storage, calibration, and diagnostics. Second, CRR is not adversarially robust: label-aware PCA-residual FGSM tests show severe collapse and objective sensitivity. Third, conformal reliability depends on whether the few normal support images are representative of future normal data. If support normals are not exchangeable with test normals, p-values can become overconfident; the uniformity tests in Section 5.4 quantify exactly this failure. Fourth, SC3R’s evidence has boundaries: the pre-registered gate passes in full at $k=4$ on all 15 MVTEC classes (seeds 0–2), replicates at $k=8$ with power-gain intervals excluding zero everywhere, and replicates on all 12 VisA classes with conservative cross-dataset transfer of MVTEC source archives—but the $k=8$ false-alarm budget is met only at $\alpha=0.05$ (at $\alpha=0.10$ the pooled rate is 0.121 against a 0.120 budget with no-harm 74%, concentrated in Gaussian noise and JPEG), the VisA no-harm rate dips to 78% at $\alpha=0.20$, and cross-dataset archives pay for their safety with roughly half

Table 11

Pixel-level localization and adversarial-fragility diagnostics on MVTec (means over classes and 5 seeds). Top: the low-storage subspace detector matches the controlled memory bank on pixel AUROC and AU-PRO. Bottom: a label-aware PCA-residual FGSM surrogate collapses image AUROC far below chance at small ϵ (score inversion), and the drop is non-monotonic in ϵ —evidence of objective sensitivity that we report as a diagnostic, not as a robustness claim.

<i>Pixel localization</i>		pixel AUROC	AU-PRO
k	Method		
1	CRR / CalibSubspace	0.944	0.820
	Controlled DINOv2 NN	0.940	0.845
4	CRR / CalibSubspace	0.957	0.862
	Controlled DINOv2 NN	0.951	0.862
8	CRR / CalibSubspace	0.960	0.876
	Controlled DINOv2 NN	0.952	0.863
<i>FGSM fragility diagnostic (image AUROC)</i>			
ϵ	k	clean $\rightarrow$ FGSM	rel. drop
2/255	4	0.937 $\rightarrow$ 0.158	83%
	8	0.945 $\rightarrow$ 0.175	82%
8/255	4	0.937 $\rightarrow$ 0.446	52%
	8	0.945 $\rightarrow$ 0.446	53%

the within-dataset power. SC3R also requires a multi-category deployment, and its matched-condition mode assumes the corruption condition is identifiable at test time. Fifth, ECE on conformal-derived scores is strongly prevalence-sensitive, so calibration numbers on anomaly-balanced test sets must not be read as deployment reliability; we treat operational false-alarm, power, and precision metrics as primary.

7. Conclusion and future work

We introduced Conformal Reliability Routing, a decoupled framework for low-storage few-shot industrial anomaly detection. Its central contribution is a pair of problem and mechanism. The problem is structural: a conformal alarm calibrated on k normal supports cannot fire below $\alpha = 1/(k+1)$, and exact discrete uniformity tests show that at its first attainable operating point the target-only rule is conservative on VisA but anti-conservative on corrupted MVTec. The mechanism is SC3R, a source-validated cross-category thresholding scheme: it passes a pre-registered gate on all 15 MVTec classes at $k=4$ —tracking nominal false-alarm rates almost exactly below and at the floor without target anomaly labels, with CI-supported power gains under every corruption condition—replicates on all 12 VisA classes, and transfers conservatively when the source archive comes from a different dataset entirely, trading power for false-alarm rates that stay under budget. Supporting these claims, CRR preserves frozen DINOv2 PCA/subspace residuals for anomaly ranking while a separate label-free reliability layer supplies decision signals, and the LOIO conformal route reduces calibration error on both full corruption benchmarks (overall ECE 0.0766 on VisA, 0.0684 on MVTec) against the complete standard toolbox—Vector and Shift-Aware Platt as well as temperature scaling, isotonic regression, and histogram binning fit on the identical calibration set—with paired Wilcoxon support in every comparison except Shift-Aware Platt at VisA $k=8$. We also report the boundaries honestly: at $k=8$ heavy-tailed corruptions erode the false-alarm budget above $\alpha=0.05$, and cross-dataset archives halve the sub-floor power. The resulting story is deliberately focused: CRR is not an MVTec AUROC SOTA or adversarial-robustness method; it is a calibrated, storage-light reliability layer for few-shot inspection systems where raw anomaly scores alone are not sufficient for deployment-quality decisions. More broadly, the results characterize a property of few-shot learned detectors themselves: what their conformal alarms can promise is bounded by an exact resolution floor, their failure direction under shift is testable, and cross-category structure the learning system already possesses is enough to restore validity below that floor.

Future work is scoped by the boundaries above: SC3R with more seeds and per-condition source validation to close the Gaussian/JPEG budget gap, and an audited vision–language baseline, which currently blocks on the absence of an official WinCLIP implementation.

Table 12

ECE comparison on the full VisA corruption benchmark. Lower is better. Bold indicates the best reliability view in each corruption cell; Figure 3 visualizes the same comparison for both benchmarks.

k	Corruption	Vector	Shift-aware	LOIO	Weighted
4	blur	0.284	0.211	0.037	0.197
4	brightness/contrast	0.285	0.212	0.036	0.199
4	gaussian noise	0.270	0.276	0.040	0.256
4	jpeg	0.288	0.230	0.043	0.210
8	blur	0.208	0.144	0.108	0.089
8	brightness/contrast	0.209	0.153	0.109	0.093
8	gaussian noise	0.190	0.191	0.117	0.189
8	jpeg	0.212	0.156	0.123	0.099

CRedit authorship contribution statement

CRedit authorship contribution statement

Given Family: Conceptualization, Methodology, Software, Validation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed during the current study are publicly available: MVTec AD (<https://www.mvtec.com/company/research/datasets/mvtec-ad>) and VisA (<https://github.com/amazon-science/spot-diff>). All derived per-image prediction tables, evaluation summaries, and sampling manifests generated during the study are available from the corresponding author on reasonable request and will be released together with the code.

Code availability

The full experimental code, including the conformal evaluation protocols, calibrator implementations, corruption pipeline, and the scripts that regenerate every table and figure in this paper, will be made publicly available upon publication.

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A. Supplementary tables

This appendix collects supporting tables whose content is summarized in the main text: the per-corruption VisA ECE breakdown (Table 12; Figure 3 visualizes the same comparison), the aggregation-sensitivity ablation (Table 13), and the novelty-positioning summary (Table 14).

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Table 13

Aggregation-sensitivity ablation: clean image AUROC (mean over representative classes, seeds 0–2) when patch residuals are pooled by max versus top- ρ mean. Bold marks the best aggregator per row; the max/top-1% choices used in the paper are within 0.01–0.02 AUROC of the per-row optimum.

dataset	k	max	top-0.5%	top-1%	top-2%	top-5%
MVTec	4	0.9571	0.9638	0.9667	0.9693	0.9621
MVTec	8	0.9578	0.9685	0.9740	0.9759	0.9727
VisA	4	0.9039	0.9101	0.9038	0.8944	0.8806
VisA	8	0.9101	0.9200	0.9147	0.9071	0.8941

Table 14

Novelty positioning: each ingredient has established prior art, and this paper's contribution begins after it. The right column marks where prior work ends and our claims start.

Work family	Established prior art	Our contribution starts after it
AnomalyDINO	DINOv2 few-shot patch/memory-bank AD	Reliability of the alarms such detectors raise
SubspaceAD	DINOv2 PCA/subspace residual ranking	Everything downstream of the residual
Calibration/robustness studies	ECE, Platt scaling, FGSM fragility	Conformal-native audit; full calibrator toolbox under one label-free protocol
Conformal AD	P-values from nonconformity scores	The few-shot floor $1/(k+1)$ and source-validated thresholding below it
SAGE-style routing	Gated shared/specialized experts	Safe-anchor routing for reliability, evaluated no-label

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